

[**N.B.** – The figures of the right margin indicate full marks. Read the stems carefully and answer the associated questions. Answer any **FIVE** questions taking at least two questions from each group]

Group - A

1. Number of accidents occurred in two different cities A and B are denoted by two variables x and y, respectively.

| City | Day 1 | Day 2 | Day 3 |
|------|-------|-------|-------|
| A    | 3     | 4     | 2     |
| B    | 5     | 3     | 2     |

An analyst observed that  $\left(\sum_{i=1}^3 x_i\right)^2 \neq \sum_{i=1}^3 x_i^2$  and  $\left(\sum_{i=1}^3 y_i\right)^2 \neq \sum_{i=1}^3 y_i^2$

- (a) What is a quantitative variable?1
- (b) Distinguish qualitative and quantitative variables.2
- (c) Show that

$$\sum_{i=1}^3 (x_i + y_i)^2 = \sum_{i=1}^3 x_i^2 + 2 \sum_{i=1}^3 x_i y_i + \sum_{i=1}^3 y_i^2$$

- 3
- (d) Evaluate the analyst’s comment.4

2. The test scores (out of 100) of 20 students in a mathematics exam were recorded as follows:

65, 72, 78, 81, 55, 88, 92, 47, 69, 73  
84, 91, 77, 63, 58, 71, 94, 86, 68, 79

- (a) What are discrete class intervals?1
- (b) What are the limitations of using discrete class intervals with continuous data?2
- (c) Make a frequency distribution and explain.3
- (d) Construct a histogram for the given data and interpret its shape.4

3. Two schools, X and Y, conducted the same mathematics examination for their students. The summary of the test scores (out of 100) is shown below:

| School | Mean Score | Standard Deviation | Number of Students |
|--------|------------|--------------------|--------------------|
| X      | 72.5       | 8.2                | 150                |
| Y      | 74.0       | 6.5                | 120                |

- (a) When is variance less than standard deviation?1
- (b) Is coefficient of variation a pure number? Explain2
- (c) Determine the combined standard deviation of the scores of the schools.3
- (d) Which school has more consistent scores? Analyze.4

Group - B

4. The first four moments around 4 of productions of a company over four years are -1.5, 17, -30, and 108.

- (a) What is raw moment?1

- (b) Can the second central moment be negative? 2
  - (c) Determine the second and third central moments. 3
  - (d) What kind of kurtosis do the data have? 4
5. The following table shows the exam scores of 10 students and the number of hours they studied for the exam. It is hypothesized that the exam score depends on the number of hours studied.

|                   |    |    |    |    |    |    |    |    |    |    |
|-------------------|----|----|----|----|----|----|----|----|----|----|
| Hours Studied (X) | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  | 10 | 11 |
| Exam Score (Y)    | 65 | 70 | 72 | 78 | 80 | 85 | 88 | 92 | 95 | 98 |

- (a) What is a scatter plot? 1
  - (b) Differentiate between the values of correlation  $r_1 = +1$  and  $r_2 = -1$  2
  - (c) Make a scatter plot using the data and state the potential direction and strength of the relationship between the variables. 3
  - (d) Find the correlation and interpret. 4
6. The monthly sales (in units) of laptops at a major electronics store are recorded below:

| Month | Jan | Feb | Mar | Apr | May | Jun | Jul | Aug |
|-------|-----|-----|-----|-----|-----|-----|-----|-----|
| Sales | 120 | 150 | 130 | 170 | 160 | 180 | 200 | 210 |

- (a) What is an irregular variation? 1
- (b) What are the components of Time Series? 2
- (c) Illustrate the trend graphically and estimate the expected laptop sales for September. 3
- (d) Compute the trend using a three-monthly moving average method and extrapolate to September, and compare with the graphical method. 4